

The double HGLM

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HGLM course at the Roslin Institute

A heteroscedastic linear model

Consider a linear model with only fixed effects both in the mean and dispersion parts:

$$\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \exp(\mathbf{X}_d\boldsymbol{\beta}_d))$$

Can also be written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$$

$$\mathbf{e} \sim N(0, \sigma_e^2)$$

$$\log(\sigma_e^2) = \mathbf{X}_d\boldsymbol{\beta}_d$$

How can this model be fitted?

GLM estimation approach

- Iterate between a linear model and a GLM
 - Estimate $\boldsymbol{\beta}$ for a given residual variance
 - Estimate $\boldsymbol{\beta}_d$: Fit $\frac{(\hat{e}_i)^2}{1-h_{ii}}$ as response variable in a gamma GLM (log link) with linear predictor $\mathbf{X}_d\boldsymbol{\beta}_d$

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DHGLM

Model with both fixed and random effects, both in the mean and dispersion parts:

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DHGLM estimation

Re-write the model as an augmented weighted linear model

$$y_a = \mathbf{T}\delta + e_a$$

$$\text{where } y_a = \begin{pmatrix} y \\ \mathbf{0} \end{pmatrix}, \mathbf{T} = \begin{pmatrix} \mathbf{X} & \mathbf{Z} \\ \mathbf{0} & \mathbf{I}_q \end{pmatrix}, \delta = \begin{pmatrix} b \\ u \end{pmatrix}, e_a = \begin{pmatrix} e \\ -u \end{pmatrix}$$

$$\text{with weights } W = \begin{pmatrix} \mathbf{I}_n \frac{1}{\sigma_e^2} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_q \frac{1}{\sigma_u^2} \end{pmatrix} \text{ giving estimates of } \beta \text{ and } u \text{ from}$$

$$\mathbf{T}'W\mathbf{T}\hat{\delta} = \mathbf{T}'W y_a$$

σ_e^2 is estimated by applying a gamma GLM to the response $\hat{e}_i^2/(1 - h_{ii})$ and linear predictor $\mathbf{X}_d\beta_d + \mathbf{Z}_d u_d$.

Use the same augmentation trick once more for treating u_d as random.

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R-ASReml code

```
###Analysis
#DHGLM calculations
mean.w <- c(rep(1,N)); res.var = 0; conv.crit = 0.00001; max.iter = 20
i.iter = 0
while (i.iter<max.iter & abs(res.var-1)>conv.crit) {
  i.iter=i.iter+1
  mean.model <- asreml(y~x,random = ~clust, weights=mean.w)
  res.var <- mean.model$sigma2 #Residual variance
  res <- resid(mean.model, type = "response")
  asreml.hv <- mean.model$hat #strange definition of hat values in ASReml
  hv.true <- asreml.hv*mean.w/res.var #Correct hat values
  y_d<-(res^2)/(1-hv.true) #Response for variance model
  var.w <- (1-hv.true)/2 #Weights for variance model
  var.model <- asreml(y_d~x,random = ~clust, weights=var.w,
    family=asreml.Gamma(link=log))
  mean.w <- 1/fitted(var.model) #Update weights for mean model
}
```


A more advanced DHGLM

Model with both fixed and random effects, both in the mean and dispersion parts:

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$$\rho = \text{cor}(u, u_d)$$

How can this model be fitted?

Felleki et al. (2012) *Genetics Research* 94:307-317.

Describes the algorithm and explains how it is connected to the h-likelihood.

Evaluated on Danish pig litter size data.

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Iterate by updating a bivariate model

Consider the bivariate linear mixed model

$$\begin{pmatrix} y \\ z_d \end{pmatrix} = \begin{pmatrix} X & 0 \\ 0 & X_d \end{pmatrix} \begin{pmatrix} \beta \\ \beta_d \end{pmatrix} + \begin{pmatrix} Z & 0 \\ 0 & Z \end{pmatrix} \begin{pmatrix} u \\ u_d \end{pmatrix} + \begin{pmatrix} e \\ e_d \end{pmatrix}$$

with working weight matrix and residual variance

$$\Sigma^{-1} = \begin{pmatrix} \Phi^{-1} & 0 \\ 0 & \text{diag}(\frac{1-q}{2}) \end{pmatrix}, \quad \text{var} \begin{pmatrix} e \\ e_d \end{pmatrix} = \Sigma \begin{pmatrix} \sigma^2 I_n & 0 \\ 0 & \sigma_d^2 I_n \end{pmatrix}$$

Here

$$\phi = V(e),$$

$z_{d,i} = \log \phi_i + \phi_i^{-1} \left(\frac{\hat{e}_i^2}{1-q_i} - \phi_i \right)$ is the linearized response (used in GLM estimation),
and q is the vector of hat values.

(σ^2 and σ_d^2 are dummy parameters used in the estimation algorithm)

Fitting algorithm

- ① Initialize weights $\Sigma^{-1} = \begin{pmatrix} \Phi^{-1} & 0 \\ 0 & \text{diag}(\frac{1-q}{2}) \end{pmatrix}$ and z_d .
- ② Fit the bivariate model with correlation structure $G \otimes A$ between a and a_d
- ③ Update $\hat{\sigma}^2$, $\hat{e}$, and q , and thereby
 $z_d = \log(\hat{\sigma}^2 \phi) + \frac{1}{\hat{\sigma}^2} \Phi^{-1}(\hat{e}^2 / (1 - q) - \hat{\sigma}^2 \phi)$, and $\text{diag}((1 - q)/2)$ in Σ^{-1} .
- ④ Repeat step 2.
- ⑤ Update $\Phi^{-1} = \text{diag}(\exp(\hat{z}_d))^{-1}$, and update Σ^{-1} .
- ⑥ Iterate steps 2-5 until convergence ($\hat{\sigma}^2 = 1$).

Simulation study

Table 3: Mean (a) and standard errors (b) of estimates of parameters for simulated data over Model III (first section) and IV (second section). The left hand column contains the simulated data structure.

(a)	Mean model		Residual variance model				Cor	
	Variances		Fixed effects			Variances		ρ
	σ_a^2	σ_p^2	β_{d0}	β_{dins}	β_{dpar}	$\sigma_{a,d}^2$	$\sigma_{p,d}^2$	
True values (Model III)	1.62	0.60	1.77	-0.17	0.35	0.09		-0.62
Original parity distribution	1.58	0.45	1.74	-0.17	0.33	0.11		-0.52
Four parities	1.59	0.61	1.73	-0.17	0.34	0.12		-0.56
Nine parities	1.63	0.60	1.74	-0.17	0.35	0.11		-0.60
True values (Model IV)	1.62	0.60	1.77	-0.17	0.35	0.09	0.06	-0.62
Original parity distribution*	1.56	0.24	1.73	-0.16	0.31	0.08	0.13	-0.61
Four parities	1.65	0.51	1.66	-0.16	0.34	0.09	0.15	-0.64
Nine parities	1.62	0.60	1.71	-0.17	0.35	0.09	0.09	-0.64

(b)								
(Model III)								
Original parity distribution	0.018	0.016	0.007	0.003	0.004	0.002		0.008
Four parities	0.015	0.012	0.005	0.003	0.004	0.002		0.007
Nine parities	0.012	0.008	0.005	0.001	0.003	0.001		0.005
(Model IV)								
Original parity distribution*	0.017	0.016	0.006	0.004	0.004	0.003	0.004	0.012
Four parities	0.017	0.012	0.007	0.003	0.004	0.002	0.003	0.008
Nine parities	0.013	0.008	0.005	0.002	0.003	0.001	0.001	0.005

* 27 out of 100 replicates did not converge. Estimates are for all replicates (with minor differences in results if these 27 replicates were included or not).